Arizona State University

Modeling and Predicting Climate Risk Using Regression Analysis

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1. Introduction

Climate change presents increasing environmental, economic, and societal risks across the globe. Understanding the factors that contribute to climate risk is essential for informed decision-making and policy planning. This project applies regression analysis techniques to model and predict climate risk using a structured panel dataset consisting of country-level climate and environmental indicators over time. The objective is to identify key drivers of climate risk and evaluate the predictive performance of statistical models.

2. Dataset

2.1 Dataset Context

The dataset used in this project is a Climate Change and Global Warming dataset obtained from Kaggle. It was created to support students, beginners, and early researchers interested in learning about climate change, global warming, and environmental impacts using data analysis and machine learning techniques. The dataset emphasizes simplicity, clarity, and usability, making it appropriate for educational and exploratory research purposes.

The dataset can be accessed at:

https://www.kaggle.com/datasets/algozee/climate-cahnge/data

2.2 Dataset Content

The dataset contains over 1200 observations and 20 features representing country-level climate and environmental indicators across multiple years. The observations span from the early 1980s through 2024 and include multiple countries such as the United States, France, India, China, Canada, Australia, and others.

Each row represents a country–year pair, forming a panel dataset (country × year) structure that allows for both cross-sectional and time-series analysis.

2.3 Column Description

The dataset contains 20 variables, including:

• year – Year in which the data record is observed
• country – Country associated with the data record

The remaining variables are divided by four categories:

Climate Variables

• global_avg_temperature – Average global surface temperature (°C)
• temperature_anomaly – Difference between observed temperature and long-term average (°C)
• max_temperature – Maximum recorded temperature for the year (°C)
• min_temperature – Minimum recorded temperature for the year (°C)
• sea_surface_temperature – Average ocean surface temperature (°C)
• sea_level_rise_mm – Average sea level rise (mm)
• annual_rainfall_mm – Total annual rainfall (mm)

Environmental Stress & Events

• heatwave_days – Number of heatwave days in a year
• drought_index – Drought severity index (higher values indicate more severe conditions)
• flood_events_count – Number of flood events recorded in a year
• climate_risk_index – Composite index (0–100) representing overall climate-related risk

Human Activity & Mitigation Indicators

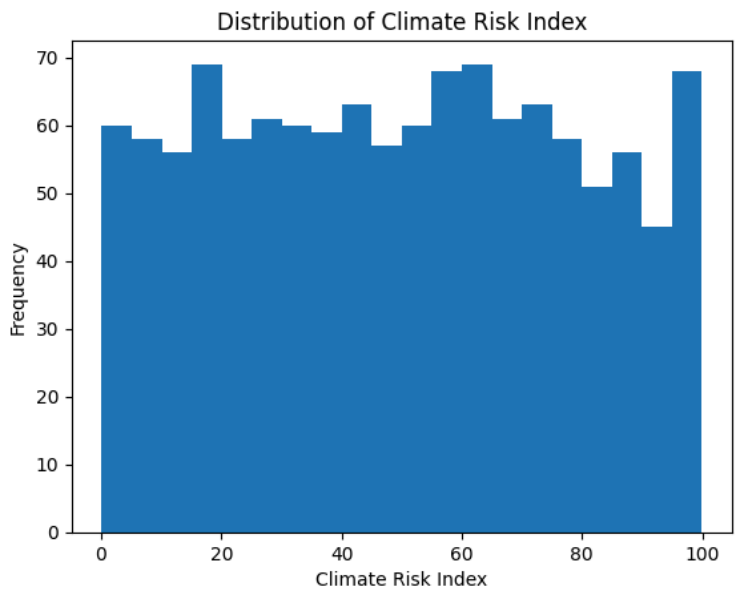
• co2_concentration_ppm – Atmospheric CO_2 concentration (ppm)
• fossil_fuel_consumption – Percentage of energy consumption from fossil fuels
• renewable_energy_share – Percentage of energy generated from renewable sources
• forest_cover_percent – Percentage of land covered by forests
• deforestation_rate – Annual rate of forest loss (%)
• air_quality_index – Air pollution index (higher values indicate poorer air quality)

Future-Oriented Variable

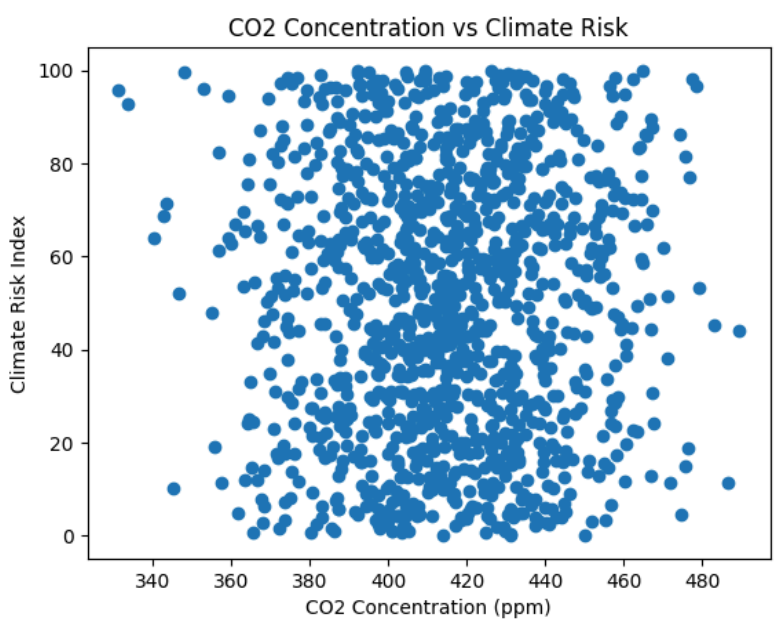
• predicted_temperature_2050 – Projected average temperature increase by the year 2050 (°C)

The climate risk index is a composite measure ranging from 0 to 100, where higher values indicate greater relative exposure and vulnerability to climate-related hazards. It serves as a summary indicator rather than a direct physical measurement and is used as the response variable in this regression analysis.

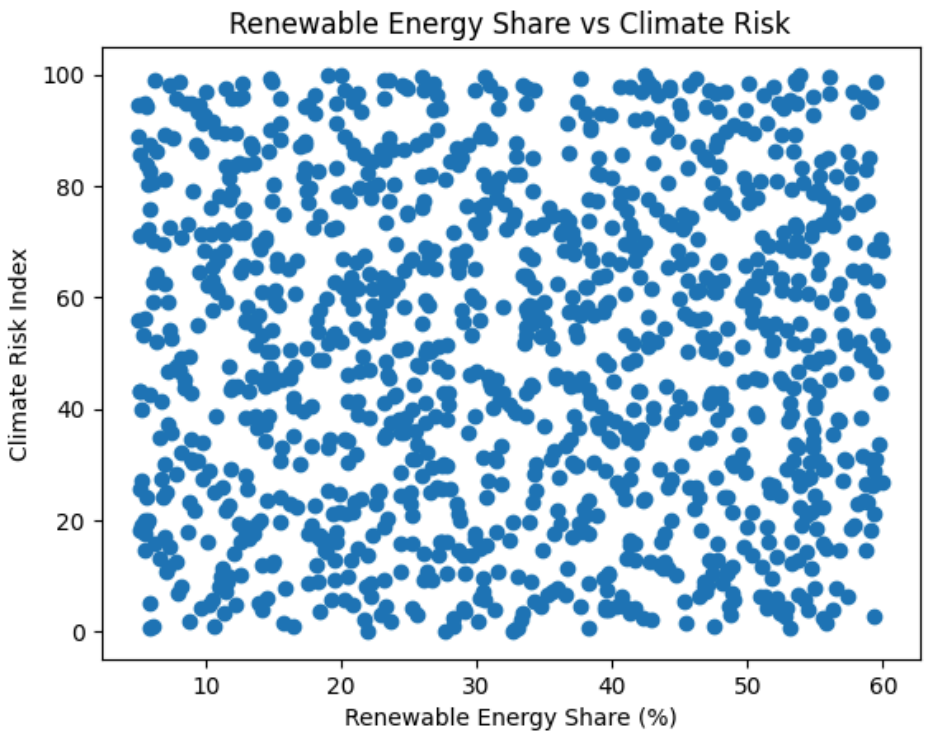
2.4 Preliminary Exploratory Analysis



The histogram illustrates the distribution of the climate risk index across all country–year observations. The values appear relatively evenly distributed across the 0–100 range, with no extreme skewness or heavy concentration in a specific interval. This suggests substantial variability in climate risk across observations, which is desirable for regression modeling. The relatively broad spread of the response variable indicates that the dataset contains sufficient variation to support predictive analysis.



The scatter plot between CO_2 concentration and climate risk index does not reveal a strong or clearly defined linear relationship. The points appear widely dispersed across the range of CO_2 values, suggesting that CO_2 concentration alone may not fully explain variation in climate risk. However, this does not eliminate its potential contribution within a multivariate regression framework. Further statistical analysis will be required to determine whether CO_2 concentration has a significant effect when considered alongside other predictors.



The relationship between renewable energy share and climate risk index appears similarly dispersed, with no immediately observable strong linear trend. Climate risk values span a wide range across different levels of renewable energy adoption. This suggests that renewable energy share alone may not determine climate risk but could contribute as part of a broader set of explanatory variables. Multivariate modeling will allow for a more rigorous assessment of its influence.

Overall, the exploratory analysis indicates substantial variability in the climate risk index and suggests that no single predictor exhibits a dominant linear relationship with climate risk in isolation. These observations support the use of multiple regression techniques to evaluate the combined effect of several predictors simultaneously.

3. Project Outline

Based on the exploratory analysis and dataset structure, a regression-based modeling framework is proposed to quantify the relationship between climate risk and explanatory variables

3.1 Research Objective

The primary objective of this project is to model and predict the **climate_risk_index** using climate, environmental, and human activity indicators available in the dataset. By applying regression analysis techniques, we aim to identify and quantify the key drivers of climate risk across countries and over time, and evaluate the predictive performance of the proposed models.

3.2 Response Variable

The response (dependent) variable for this analysis is the **climate_risk_index**, a composite measure ranging from 0 to 100 that represents the relative level of climate-related risk for a country in a given year. Higher values indicate greater exposure and vulnerability to climate-related hazards.

3.3 Predictor Variables

The predictor (independent) variables considered include:

- heatwave_days
- drought_index
- flood_events_count
- deforestation_rate
- fossil_fuel_consumption
- co2_concentration_ppm
- renewable_energy_share
- forest_cover_percent
- air_quality_index

Additional climate variables may be considered if appropriate. Variable selection techniques will be applied to determine the most statistically significant predictors for the final model.

3.4 Proposed Methodology

The following steps will be implemented:

1. Exploratory Data Analysis (EDA)
Initial data exploration will be conducted to examine distributions, identify trends, detect outliers, and assess relationships between predictor variables and climate risk.

2. Multiple Linear Regression
Multiple linear regression will be used as the primary modeling framework due to the continuous nature of the response variable. This approach enables estimation and interpretation of the relationship between climate risk and multiple explanatory variables. Statistical significance will be evaluated at conventional confidence levels.

3. Variable Selection
Techniques such as stepwise selection, adjusted R^2 comparison, and statistical significance testing will be applied to determine the most appropriate subset of predictors.

4. Model Diagnostics
Model assumptions including linearity, homoscedasticity, normality of residuals, and multicollinearity will be evaluated to ensure validity of the regression model.

5. Model Comparison
Alternative modeling approaches may be explored and compared based on predictive performance.

6. Prediction and Validation
The dataset will be divided into training and testing sets. Model performance will be evaluated using metrics such as Mean Squared Error (MSE), Mean Absolute Error (MAE), and R^2 by comparing predicted climate risk values to observed values.